

BERDO Retrofit Prioritization Dashboard

Final Project Write-Up

MISM6210 - Information Visualization & Dashboard Design

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1. The Research Problem

Boston's Building Emissions Reduction and Disclosure Ordinance (BERDO) is one of the most significant climate policies at the city level in the United States. It requires all large buildings—those above 20,000 square feet to progressively reduce their greenhouse gas emissions over time. Buildings are responsible for roughly 70% of Boston's total emissions, which makes this ordinance the city's single most important tool for reaching its 2050 carbon neutrality target. Non-compliant buildings face financial penalties and public disclosure.

The challenge is one of scale and prioritisation. The BERDO portfolio contains 3,271 large buildings spread across 16 property types and dozens of zip codes. Not every building needs the same level of intervention. With limited retrofit budgets and workforce capacity, the city cannot address all buildings at once. The practical question is: ***which specific buildings should Boston prioritise for energy retrofits to achieve the greatest reduction in emissions and the highest cost savings?***

This project set out to answer that question using visual analytics. Our goal was to build a dashboard system that a decision-maker whether in the city's Environment Department or a building portfolio manager—could use to identify the highest-impact retrofit targets, understand their geographic distribution, and plan a phased intervention strategy.

2. Data and Methodology

2.1 Dataset and Cleaning

We used a publicly available BERDO reporting dataset from the year 2025, containing building-level energy and emissions data for Boston's large commercial and residential properties. The original dataset contained missing values in several key fields (such as Energy Star Score, natural gas and electricity usage), duplicate building entries, and inconsistent formatting across property type labels. We performed data cleaning and preparation using Python in a Jupyter Notebook environment. This included removing duplicate records, imputing or dropping rows with missing values in critical fields, standardising property type labels, and ensuring all numeric columns were in the correct data types. After cleaning, the final dataset contained 3,271 complete records with 29 attributes and zero null values. This cleaned CSV was then connected to Tableau for analysis.

The 29 attributes cover building identifiers (BERDO ID, address, zip code, owner, property type), energy metrics (total site energy, Site EUI, natural gas and electricity usage), emissions metrics (GHG emissions in kgCO₂e and metric tons, emissions intensity), building characteristics (floor area, size category, Energy Star score, water usage), compliance information (reporting status, compliance risk, projected compliance year), and six green technology flags (EV charging, battery storage, backup generator, PPA, REC, community choice electricity).

2.2 Analytical Framework

Opportunity Score: We created a percentile-based composite metric combining Site EUI, emissions intensity, floor area, and Energy Star score. Each building receives a score from 0 to 100, where higher scores indicate greater retrofit potential.

3-Tier Classification: Based on the Opportunity Score, buildings were classified into three tiers. Tier 1 (Targeted Retrofits) includes the top 10%—327 buildings. Tier 2 (Efficiency Upgrades) covers the next 20%—654 buildings. Tier 3 (Monitor) includes the remaining 70%—2,290 buildings.

K-Means Clustering: To validate our expert-defined tiers, we applied Tableau’s built-in k-means clustering (k=4) on the scatter plot of Floor Area vs. Site EUI. The statistical clusters aligned closely with our tier assignments, confirming that the tiering reflects the natural structure of the data.

2.3 Calculated Fields

We created over 15 calculated fields in Tableau before building any worksheets. These included tier ordering and labelling fields (Tier Order, Tier Label Short, Risk Order), a compliance year bucket formula to group buildings into “Already Due” and “2026–2030” categories, energy unit conversions (converting electricity from kWh to kBtu using the 3.412 factor for comparable energy source analysis), gas and electricity percentage fields for the stacked bar, risk weight encoding for the heatmap, green technology adoption flags (using 1.0/0.0 decimal output to force Tableau to treat them as measures), formatted KPI strings, and a boolean Is Tier 1 filter field. Building these upfront meant each worksheet could be assembled quickly by dragging pre-built measures.

3. Visualisations Produced

We built 16 Tableau worksheets organised across 4 dashboards. The dashboards follow a zoom-in narrative: city-wide overview, sector analysis, building-level targets, actionable retrofit plan. Each chart type was chosen specifically for the analytical question it needed to answer.

Dashboard 1 — City-Wide Energy Pulse (9 worksheets)

Question: *How big is Boston’s building emissions challenge?*

Five KPI cards display the portfolio’s scale: 3,271 buildings, 1,446K metric tons CO₂e, \$763M annual energy cost, 77.6 average EUI, and 327 Tier 1 buildings. We used large standalone numbers rather than charts because KPIs need to be absorbed in seconds—the audience grasps the scale immediately without reading axes or legends.

A **treemap** was chosen over a bar chart because it shows part-to-whole relationships in two dimensions simultaneously—rectangle size encodes total GHG emissions while colour encodes average EUI. This dual encoding immediately reveals that the biggest emitters (Multifamily Housing, Other) are not the most energy-wasteful per square foot—that distinction belongs to Laboratory at 222.5 kBtu/ft². A bar chart could only show one of these dimensions at a time.

A **donut chart** shows the 10/20/70 tier split (327 / 654 / 2,290). We chose a donut over a pie chart because the centre space displays the total building count (3,271), and over a bar chart because proportional splits are more intuitive as segments of a whole. A **compliance risk heatmap** was chosen because it cross-references two categorical dimensions (property type × risk level) in a matrix format—something a bar chart cannot do without becoming cluttered. This reveals that “Other” has 22 high-risk buildings, the most of any sector.

The **compliance timeline** (stacked bar with year buckets) adds the time dimension that the other charts lack. It answers “when is compliance due?” and reveals that all 327 Tier 1 buildings have already passed their deadline—transforming the retrofit argument from future planning to current regulatory urgency.

Dashboard 2 — Sector Energy Benchmarking (2 worksheets)

Question: *Which sectors consume the most energy, and how much do buildings vary within each?*

A **lollipop chart** was chosen over a standard bar chart because the thin stem with a circle endpoint reduces visual clutter while drawing the eye to the actual value. A colour gradient on the circles encodes EUI intensity, adding a second layer of information. This ranks all 16 property types by average EUI with a portfolio average reference line at 77.6 kBtu/ft². Supermarkets (227.0) and Laboratories (222.5) sit nearly 3x above this line.

A **box-and-whisker plot** was chosen because it reveals what averages hide—the median, quartiles, spread, and outliers within each sector. A simple bar of averages would show Multifamily Housing at 69.2 kBtu/ft² and appear acceptable, but the box plot reveals outlier buildings reaching 450+ kBtu/ft². This finding directly motivated the need for building-level analysis in Dashboard 3.

Dashboard 3 — Building Analysis & Energy Profile (2 worksheets)

Question: *Which individual buildings are the highest-impact targets, and how dependent are sectors on natural gas?*

The centrepiece is a **quadrant scatter plot** with Floor Area on the x-axis and Site EUI on the y-axis. We chose a scatter plot because it is the only chart type that shows the relationship between two continuous variables at the individual building level. Median reference lines divide the chart into four quadrants—“HIGH IMPACT TARGETS,” “Small Wasters,” “Efficient Giants,” and “Low Priority”—creating a visual decision framework. Colour encodes tier, shape encodes k-means cluster membership (4 clusters), and size encodes GHG emissions. This triple encoding lets the viewer compare expert judgement with statistical grouping on a single chart. The top-right quadrant clearly shows Tier 1 buildings clustering where both floor area and energy intensity are high. The number one target—20 Gillette Park—has an EUI of 667.7 kBtu/ft², nearly 9x the portfolio average.

A **100% stacked bar** was chosen for the energy source mix because it shows proportional composition—the split between natural gas and electricity—for every sector on the same scale. This reveals that Manufacturing (88%), Worship Facilities (73%), and Multifamily Housing (58%) are heavily gas-dependent, meaning electrification must be central to any retrofit strategy for these sectors.

Dashboard 4 — Retrofit Priority Planner (3 worksheets)

Question: *Where should Boston deploy retrofits, what technology gaps exist, and which buildings come first?*

A **zip code symbol map** was chosen because geographic clustering is impossible to see in a table or bar chart. Bubble size encodes Tier 1 building count and colour encodes average Opportunity Score. The map reveals that the top 5 zip codes (02115, 02215, 02116, 02210, 02118) contain 156 of the 327 Tier 1 buildings—48% of all priority targets concentrated in the Longwood-Fenway-Back Bay-Seaport corridor.

A **grouped bar chart** compares green technology adoption rates across tiers for six technologies. Side-by-side bars were chosen over stacked bars because the goal is to compare Tier 1 vs. Tier 2 vs. Tier 3 adoption rates directly. This produced a surprising finding: Tier 1 buildings lead in every category (56.6% have backup generators, 10.7% have EV charging) yet remain the worst performers—demonstrating that technology additions cannot substitute for structural retrofits.

A **ranked table** lists the top 10 buildings by Opportunity Score with annual cost, EUI, floor area, and GHG emissions. We chose a table rather than a chart because this is the actionable deliverable—a decision-maker needs specific building names, addresses, and cost figures to take to a budget meeting, not abstract visual encodings.

4. Conclusions and Recommendations

First, the biggest emitters are not the most wasteful buildings. Total emissions volume and per-square-foot energy intensity tell different stories. A strategy based only on total emissions would target Multifamily Housing and “Other” properties, but the buildings with the highest waste per square foot are Laboratories, Supermarkets, and Manufacturing facilities.

Second, sector-level averages are misleading. Multifamily Housing’s average of 69.2 kBtu/ft² hides individual buildings at 450+. Building-level analysis is essential.

Third, every Tier 1 building has already passed its compliance deadline. 100% of the 327 Tier 1 buildings fall in the “Already Due” bucket. This is a current regulatory liability, not a future planning exercise.

Fourth, green technology adoption does not fix fundamental building inefficiency. Tier 1 buildings lead in adoption rates yet remain the worst performers. Deep structural retrofits (HVAC, building envelope) are required.

Fifth, geographic concentration enables efficient deployment. 48% of all Tier 1 buildings sit in just 5 zip codes in the Longwood-Fenway-Back Bay-Seaport corridor, enabling area-based retrofit programs.

Based on these findings, we recommend a three-phase approach:

Phase 1: Commission energy audits for the top 10 Tier 1 buildings, prioritise HVAC and building envelope retrofits, and allocate capital budget for buildings with over \$10M in annual energy costs.

Phase 2: Deploy area-based retrofit programs in zip codes 02115, 02215, and 02116, targeting 156 Tier 1 buildings in the top 5 zip clusters, and prioritise electrification for gas-dependent sectors.

Phase 3: Track 654 Tier 2 buildings for efficiency upgrades, monitor Tier 3 for compliance drift, and re-evaluate tiering annually as retrofits are completed.